VLSI INTERNSHIP – TASK 1

INTRODUCTION TO VLSI DESIGN FLOW

AND BASIC DIGITAL LOGIC IMPLEMENTATION

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Internship Domain : VLSI Design

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1. Introduction

Very Large-Scale Integration (VLSI) is the process of integrating millions or even billions of transistors onto a single semiconductor chip. VLSI technology is the foundation of modern electronic systems and is widely used in processors, memory devices, communication systems, embedded systems, and consumer electronics. The advancement of VLSI has enabled high-speed computation, reduced power consumption, compact device size, and improved system performance.

1.1 Objectives

• To understand the fundamentals of VLSI technology.

• To study the VLSI design flow.

• To learn the operation of basic logic gates.

• To verify logic gate functionality using truth tables.

• To design and simulate Half Adder circuits.

• To implement and verify Full Adder circuits.

1.2 Applications of VLSI

• Microprocessors

• Memory Devices

• Smartphones

• Embedded Systems

2. VLSI Design Flow

System Specification

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Architecture Design

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RTL Design

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Functional Verification

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Logic Synthesis

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Physical Design

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Timing Analysis

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Fabrication

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Testing and Validation

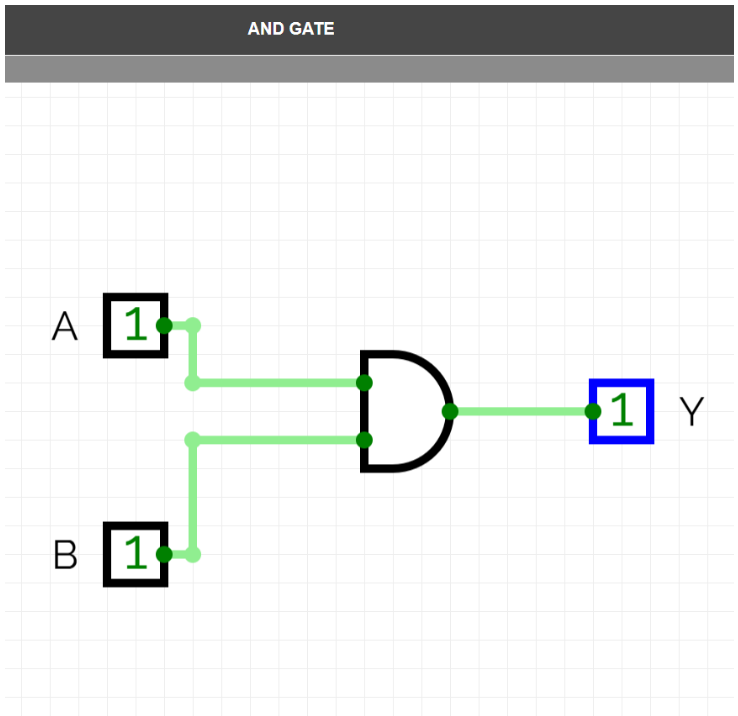
Explanation:

- **System Specification:** Defines system requirements and functionality.
- **Architecture Design:** Develops the overall structure of the system.
- **RTL Design:** Hardware description using Verilog or VHDL.
- **Functional Verification:** Ensures correct functionality through simulation.
- **Logic Synthesis:** Converts RTL code into gate-level netlist.
- **Physical Design:** Placement and routing of components.
- **Timing Analysis:** Verifies timing constraints.
- **Fabrication:** Manufacturing of the chip.
- **Testing and Validation:** Final verification of the fabricated device.

BASIC LOGIC GATES

3. AND Gate

3.1 Circuit Diagram



3.2 Description

The AND gate is a basic digital logic gate that performs logical multiplication. The output of an AND gate becomes HIGH (1) only when all the inputs are HIGH. If any one of the inputs is LOW (0), the output becomes LOW.

3.3 Boolean Expression

$$Y = A \cdot B$$

3.4 Truth Table

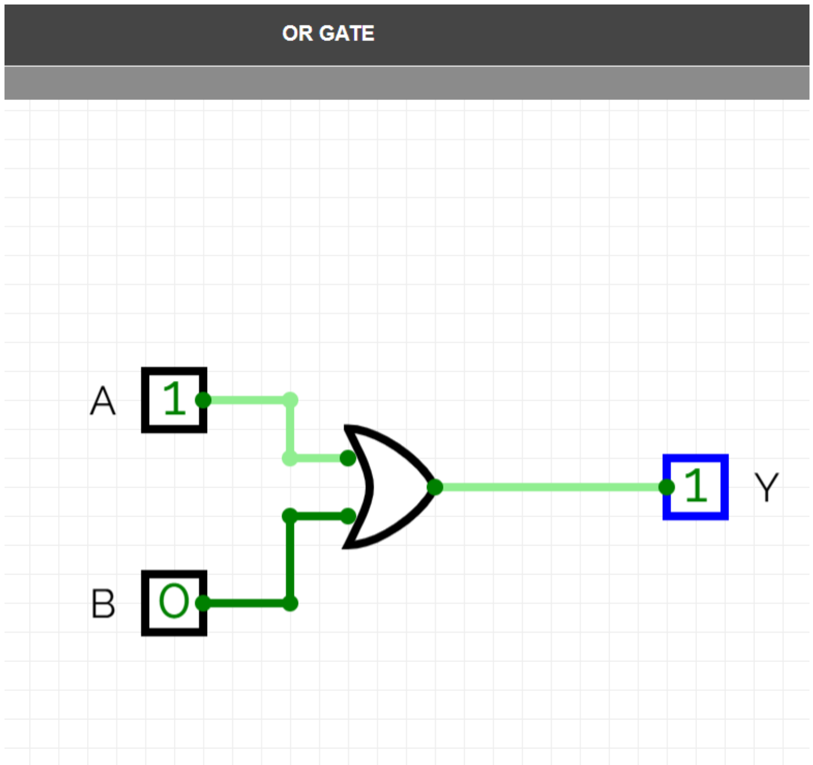
A	**B**	**Y**
0	0	0
0	1	0
1	0	0
1	1	1

3.5 Observation

- The output is HIGH only when both inputs are HIGH.

4. OR Gate

4.1 Circuit Diagram



4.2 Description

The OR gate performs logical addition. The output becomes HIGH whenever at least one input is HIGH.

4.3 Boolean Expression

$$Y = A + B$$

4.4 Truth Table

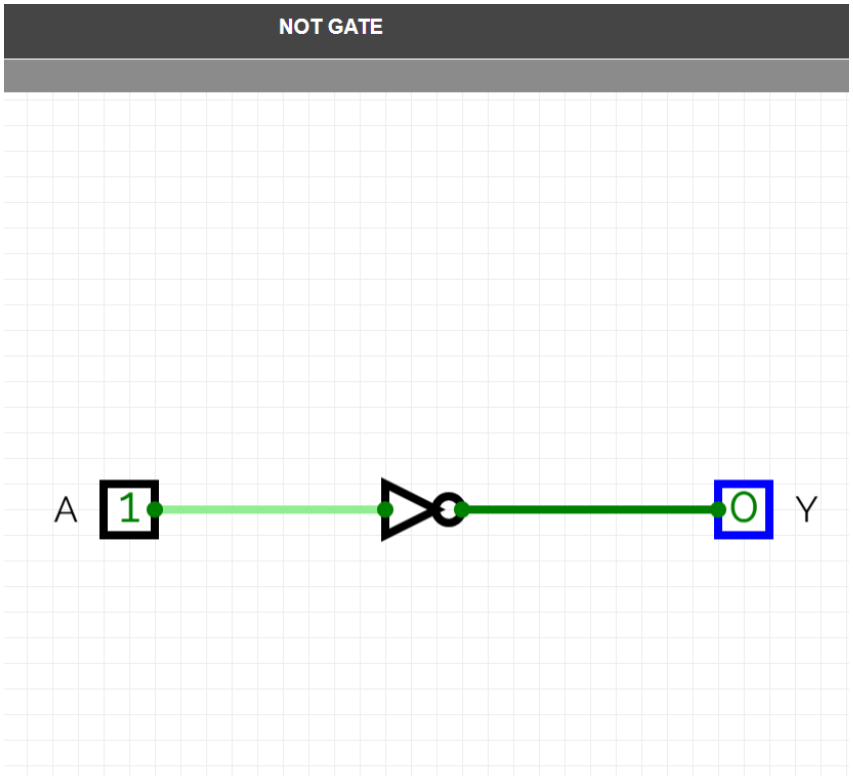
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

4.5 Observation

- The output is HIGH if one or both inputs are HIGH.

5. NOT Gate

5.1 Circuit Diagram



5.2 Description

The NOT gate is also called an inverter. It produces the complement of the input signal.

5.3 Boolean Expression

$$Y = \bar{A}$$

5.4 Truth Table

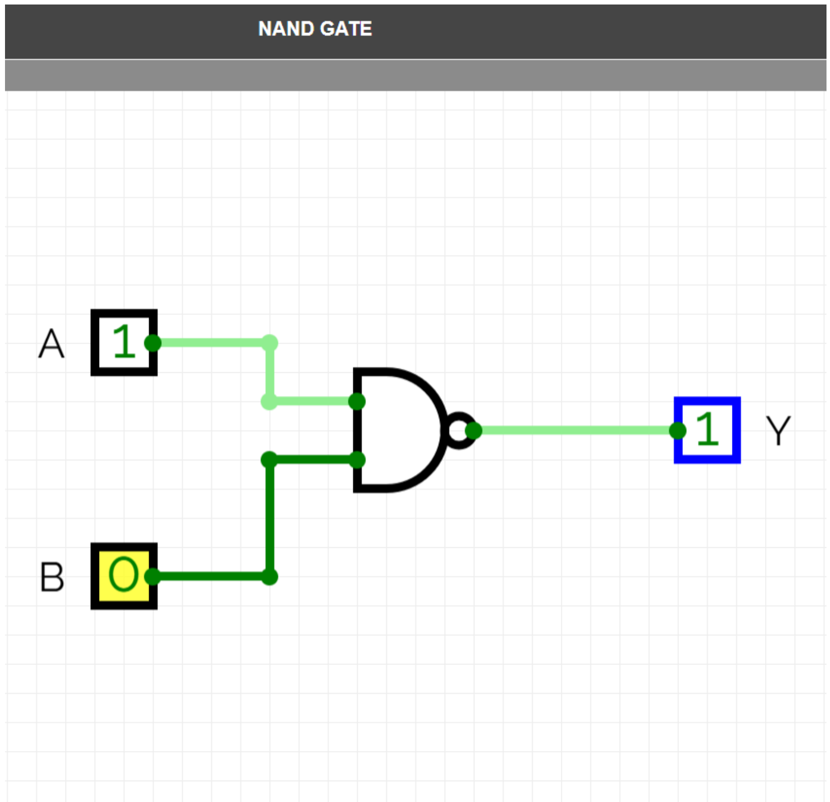
A	Y
0	1
1	0

5.5 Observation

- The output is always opposite to the input.

6. NAND Gate

6.1 Circuit Diagram



6.2 Description

The NAND gate is formed by combining an AND gate followed by a NOT gate. It is a universal gate.

6.3 Boolean Expression

$$Y = \overline{A \cdot B}$$

6.4 Truth Table

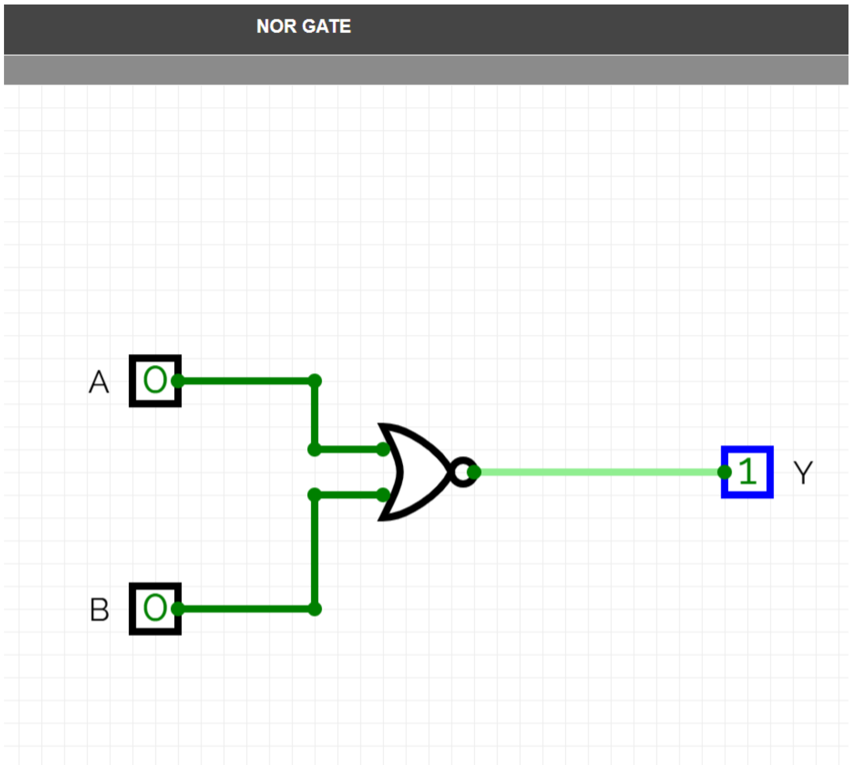
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

6.5 Observation

- The output becomes LOW only when both inputs are HIGH.

7. NOR Gate

7.1 Circuit Diagram



7.2 Description

The NOR gate is obtained by connecting a NOT gate after an OR gate. It is also a universal gate.

7.3 Boolean Expression

$$Y = A \bar{+} B$$

7.4 Truth Table

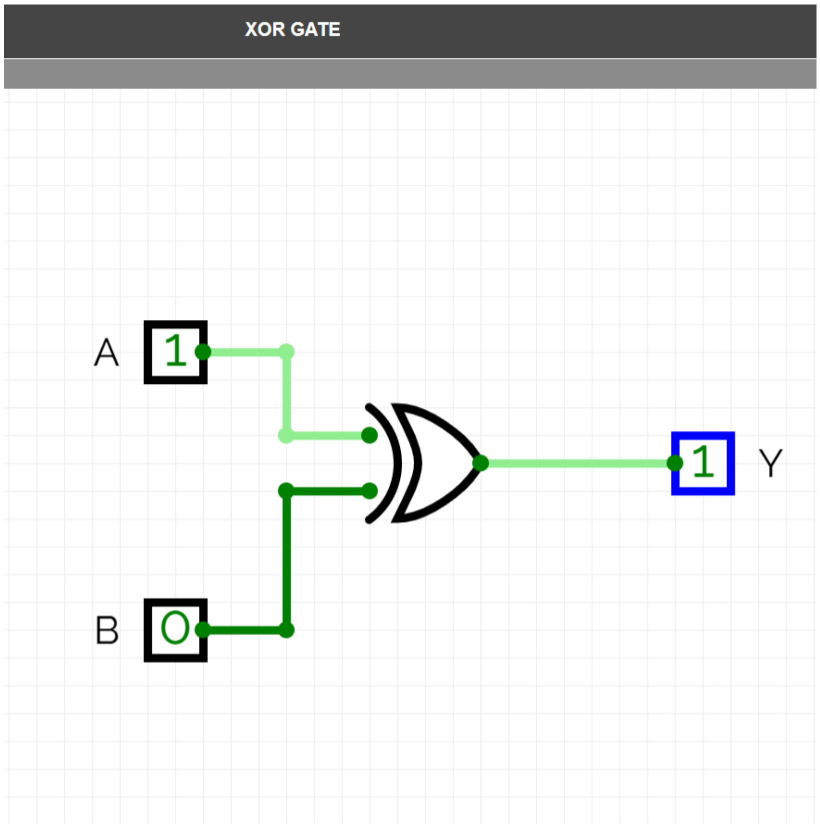
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

7.5 Observation

- The output is HIGH only when all inputs are LOW.

8. XOR Gate

8.1 Circuit Diagram



8.2 Description

The XOR gate produces a HIGH output only when the inputs are different. It is commonly used in arithmetic circuits.

8.3 Boolean Expression

$$Y = A \oplus B$$

8.4 Truth Table

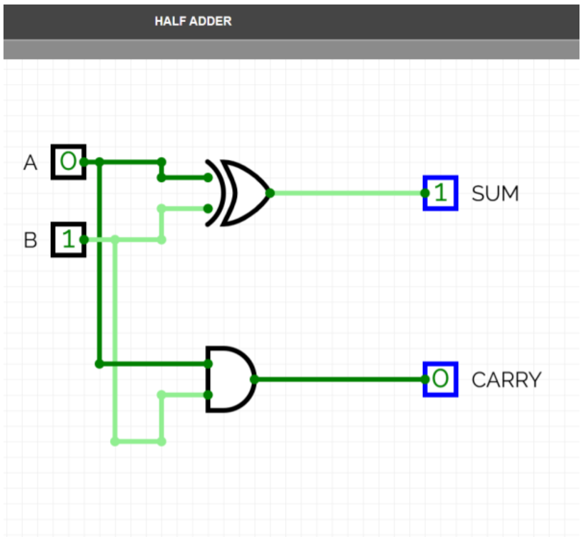
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

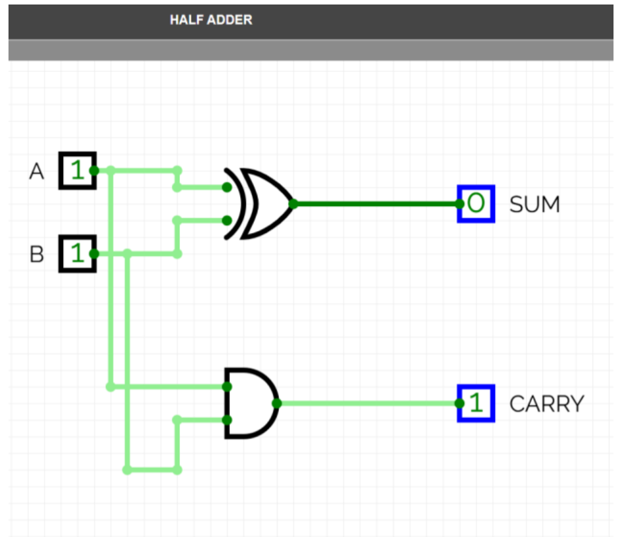
8.5 Observation

- The output becomes HIGH when the inputs are different.

9. Half Adder

9.1 Circuit Diagram





9.2 Description

A Half Adder is a combinational logic circuit used to add two single-bit binary numbers. It consists of two inputs (A and B) and produces two outputs: Sum and Carry. The Sum output is generated using an XOR gate, while the Carry output is generated using an AND gate. Half Adders are the basic building blocks of arithmetic circuits and are used in digital systems for binary addition.

9.3 Boolean Expression

Sum:

$$S = A \oplus B$$

Carry:

$$C = A \cdot B$$

Where:

- A = First Input
- B = Second Input
- S = Sum Output
- C = Carry Output

9.4 Truth Table

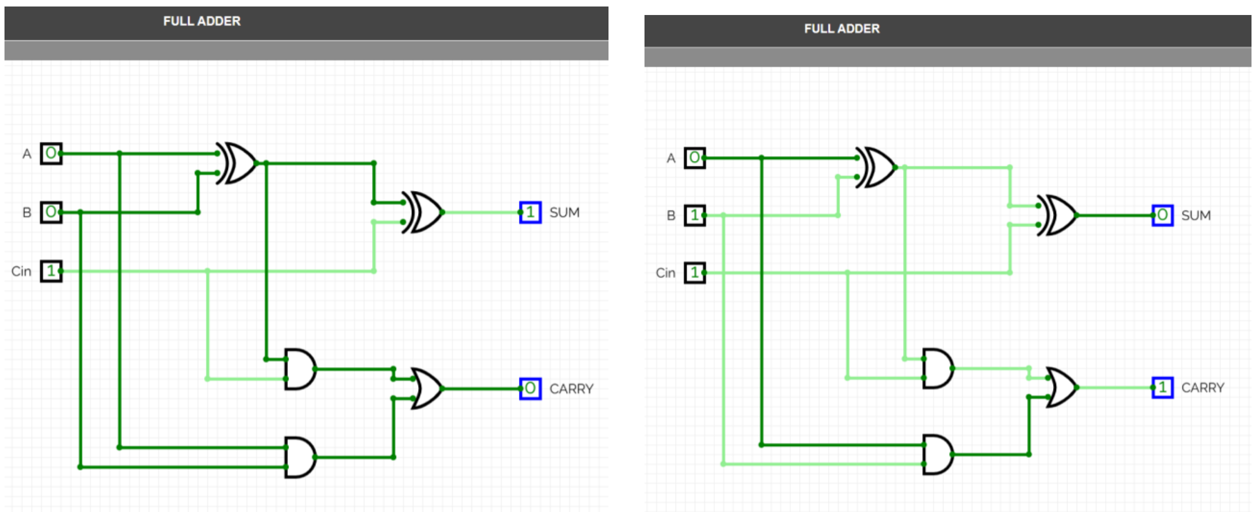
A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

9.5 Observation

- From the simulation results, it is observed that the Sum output is HIGH when the inputs are different, while the Carry output becomes HIGH only when both inputs are HIGH. The obtained outputs match the Half Adder truth table.

10. Full Adder

10.1 Circuit Diagram



10.2 Description

A Full Adder is a combinational logic circuit used to add three binary inputs: A, B, and Carry-in (Cin). It generates two outputs: Sum and Carry-out (Cout). A Full Adder can be implemented using two Half Adders and one OR gate. It is widely used in arithmetic and logic units (ALUs), processors, and digital systems for multi-bit binary addition.

10.3 Boolean Expression

Sum:

$$\boldsymbol{S = A \oplus B \oplus Cin}$$

Carry:

$$Cout = (A \cdot B) + (Cin \cdot (A \oplus B))$$

Where:

- A = First Input
- B = Second Input
- Cin = Carry Input
- S = Sum Output
- Cout = Carry Output

10.4 Truth Table

A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

10.5 Observation

- The simulation results confirm that the Full Adder correctly performs binary addition of three inputs. The Sum output follows the XOR operation of all three inputs, while the Carry output becomes HIGH whenever two or more inputs are HIGH. The outputs obtained from the simulation match the Full Adder truth table.

11.Results

The logic gates AND, OR, NOT, NAND, NOR, and XOR were successfully designed and simulated using CircuitVerse. The outputs obtained from the simulations matched the expected truth tables. Furthermore, Half Adder and Full Adder circuits were implemented using logic gates and verified for all possible input combinations. The simulation results confirmed the correctness of the designed digital circuits.

12.Applications

- **Logic Gates**

  - Used in digital electronic systems.
  - Used in processors and microcontrollers.
  - Used in control and decision-making circuits.

- **Half Adder**
  - Used for adding two binary digits.
  - Forms the building block of arithmetic circuits.
  - Used in digital calculators and processors.
- **Full Adder**
  - Used for adding three binary inputs.
  - Forms the foundation of multi-bit adders.
  - Used in Arithmetic Logic Units (ALUs) and CPUs.

13.Conclusion

The basic logic gates and combinational circuits were successfully designed and simulated using CircuitVerse. The functionality of AND, OR, NOT, NAND, NOR, XOR, Half Adder, and Full Adder circuits was verified through simulation and truth table analysis. The obtained outputs matched the theoretical results, demonstrating the correctness of the implemented digital circuits. This task enhanced the understanding of digital logic design and combinational circuit implementation.

- **References**

1. CircuitVerse Digital Logic Simulator
2. Digital Electronics – Morris Mano
3. Fundamentals of Digital Logic Design